

Green's formula + Gauss's formula

1. it is $\oint_L \frac{y dx - x dy}{2(x^2+y^2)}$, $L: x^2+y^2=2$, clockwise

Ans: $p = \frac{y}{2(x^2+y^2)}$, $q = \frac{-x}{2(x^2+y^2)}$



$\frac{\partial p}{\partial x} = \frac{x^2-y^2}{2(x^2+y^2)^2}$, $\frac{\partial q}{\partial y} = \frac{x^2-y^2}{2(x^2+y^2)^2}$

$\frac{\partial p}{\partial x} = \frac{\partial q}{\partial y}$ ($x^2+y^2 \neq 0$)

for $(0,0)$: in L is $k: \begin{cases} x = \varepsilon \cos \theta \\ y = \varepsilon \sin \theta \end{cases}$, $\theta \in [0, 2\pi]$ clockwise

$\oint_{L+L'} p dx + q dy = \iint_D (\frac{\partial p}{\partial x} - \frac{\partial q}{\partial y}) dx dy = 0$

$\Rightarrow \oint_{L'} p dx + q dy$

$\oint_{L'} = \int_0^{2\pi} \frac{-\varepsilon^2 \sin^2 \theta - \varepsilon^2 \cos^2 \theta}{2\varepsilon^4} d\theta = -\pi$

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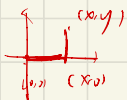
(1) $4 \sin x \sin 3y \cos x dx - 3 \cos 3y \cos 2x dy$

$\therefore p = 4 \sin x \sin 3y \cos x$

$q = -3 \cos 3y \cos 2x$

$\frac{\partial p}{\partial y} = 12 \sin x \cos x \cos 3y = 6 \cos 3y \sin 2x$

$\frac{\partial q}{\partial x} = 6 \cos 3y \sin 2x = \frac{\partial p}{\partial y}$

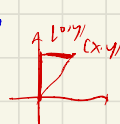


$\therefore \oint_{\partial D} u(x,y) = \int_{(0,0)}^{(x,y)} 4 \sin x \sin 3y \cos x dx - 3 \cos 3y \cos 2x dy$

$= \int_0^x 0 dx + \int_0^y -3 \cos 3y \cos 2x dy = -\cos 2x \sin 3y + C$

(2) $(3x^2y + \cos y^2) dx + (x^3 + 8x^2y + 12ye^y) dy$

$\therefore \frac{\partial p}{\partial x} = 3x^2 + \cos y^2 = \frac{\partial q}{\partial y}$



$\therefore u(x,y) = \int_{(0,0)}^{(x,y)} (3x^2y + \cos y^2) dx + (x^3 + 8x^2y + 12ye^y) dy$

$= \int_0^y 12ye^y dy + \int_0^x (3x^2y + \cos y^2) dx + C$

$= x^3y + 4x^2y^2 + 12ye^y - e^y + C$

(3) $(2x \cos y + y^2 \cos x) dx + (2y \sin x - x^2 \sin y) dy$

$\frac{\partial p}{\partial x} = 2y \cos x - 2x \sin x = \frac{\partial q}{\partial y}$

$u(x,y) = \int_0^x 2x dx + \int_0^y (2y \sin x - x^2 \sin y) dy + C$
 $= y^2 \sin x + x^2 \cos y + C$

$\therefore$ 构造函数 u , 使得在右半平面 $x > 0$

$\int_L 2xy(x^2+y^2)^{\lambda} dx - x^2(x^2+y^2)^{\lambda} dy$ 与 $\frac{\partial p}{\partial x} = \frac{\partial q}{\partial y}$

求 $u(x,y)$

$p = 2xy(x^2+y^2)^{\lambda}$, $q = -x^2(x^2+y^2)^{\lambda}$

$\frac{\partial p}{\partial x} = \frac{\partial q}{\partial y}$

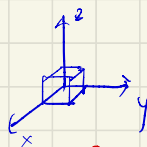
$\Rightarrow (x^2+y^2)(\lambda+1) = 0 \Rightarrow \lambda = -1$

$u(x,y) = \int_{(0,0)}^{(x,y)} \frac{2xy}{x^2+y^2} dx - \frac{x^2}{x^2+y^2} dy$

$= \int_1^x 0 dx - \int_0^y \frac{x^2}{x^2+y^2} dy = -\arctan \frac{y}{x^2}$

4. (1) $\iiint_V 4xz dy dz - y^2 dz dx + yz dx dy$

$\Sigma: x=0, y=0, z=0, x=1, y=1, z=1$ clockwise



$\oint_{\Sigma} = \iiint_V (\frac{\partial p}{\partial x} + \frac{\partial q}{\partial y} + \frac{\partial r}{\partial z}) dv = \iiint_V (4z-y) dv$

$= \int_0^1 dx \int_0^1 dy \int_0^1 (4z-y) dz = \frac{3}{2}$

$$(2) \vec{F} = \iiint_V \left(c \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} \right) dV$$

$$= \iiint_V (cx^2 + y^2 + z^2) dV$$

$$= \int_0^{2\pi} d\theta \int_0^{\frac{\pi}{2}} d\varphi \int_0^R r^2 \sin\varphi dr = \frac{2}{3} \pi R^5$$

$$(3) \vec{F} = \iiint_V \left(c \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} \right) dV = \iiint_V 3 dV = 81\pi$$

$$(4) \vec{F} \in \Sigma_1 \left\{ \begin{array}{l} z=a \\ x^2+y^2 \leq a^2 \end{array} \right. \quad \text{上底}$$

$$I = \oint_{\Sigma_1} -y dz$$

$$= \iiint_{x^2+y^2 \leq a^2, 0 \leq z \leq a} z dx dy dz - \iint_{x^2+y^2 \leq a^2} a^2 dx dy$$

$$= 2 \int_0^a dz \iint_{x^2+y^2 \leq a^2} z dx dy - 2\pi a^2 = -\frac{1}{2} 2\pi a^2$$

$$(5) \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} = 0$$

$$\Rightarrow \vec{F} \in \Sigma_1, \text{ 以 } (1,1,1) \text{ 为球心, 1 为半径的上半球面}$$

$$\Sigma_2 \text{ 介于 } x^2+y^2 \leq 1 \text{ 与 } \frac{(x-1)^2}{16} + \frac{(y-1)^2}{9} = 1 \text{ 之间, 下底}$$

$$\iint_{\Sigma_1 + \Sigma_2} \frac{-x dy dz + y dz dx + z dx dy}{r^3} = 0$$

$$\Rightarrow I_1 = \iint_{\Sigma_1} -x dy dz + y dz dx + z dx dy$$

$$\vec{F} \in \Sigma_2, z=0, x^2+y^2 \leq 1. \quad \Rightarrow I_2 = \iint_{\Sigma_2} -x dy dz + y dz dx + z dx dy = -2\pi$$

$$I = -I_1 + 0 - 0 - (-2\pi) = 2\pi.$$